

DINESH SHEELAM — AI/ML SKILL MASTERY CHECKLIST

2-Week Structured Preparation Plan • Beginner → Expert Coverage • 2025-2026 Market Aligned

■ Beginner

■ Intermediate

■ Advanced/Expert

■ = Check when confident

■ 1. PYTHON FOR AI/ML

Core Language Fundamentals

☐	Variables, Data Types (int, float, str, bool, None)	Beginner
☐	Lists, Tuples, Sets, Dictionaries — CRUD & iteration	Beginner
☐	List/Dict/Set comprehensions	Beginner
☐	Functions — args, kwargs, default params, return values	Beginner
☐	Lambda functions & map/filter/zip	Beginner
☐	Modules & packages — import, __init__.py	Beginner
☐	Exception handling — try/except/finally, custom exceptions	Beginner
☐	File I/O — open/read/write/CSV/JSON	Beginner
☐	String methods & f-strings (format, split, strip, replace)	Beginner
☐	Conditional logic & loops (for, while, break, continue)	Beginner

Object-Oriented Programming (OOP)

☐	Classes & Objects — __init__, self, attributes	Beginner
☐	Instance methods vs class methods vs static methods	Intermediate
☐	Inheritance & method overriding	Intermediate
☐	Encapsulation — public, protected, private attributes	Intermediate
☐	Polymorphism & duck typing	Intermediate
☐	Magic/Dunder methods — __str__, __repr__, __len__, __eq__	Intermediate
☐	Abstract classes (ABC) & interfaces	Intermediate
☐	Dataclasses (@dataclass decorator)	Intermediate
☐	Composition vs Inheritance patterns	Advanced
☐	Design patterns relevant to ML — Factory, Strategy, Observer	Advanced

Intermediate & Advanced Python

☐	Decorators — @staticmethod, @classmethod, custom decorators	Intermediate
☐	Generators & iterators — yield, next(), iter()	Intermediate
☐	Context managers — with statement, __enter__ / __exit__	Intermediate

☐	*args and **kwargs unpacking deeply	Intermediate
☐	Type hints — int, str, List, Dict, Optional, Union (typing module)	Intermediate
☐	Virtual environments — venv, pip, requirements.txt	Intermediate
☐	Async / Await — asyncio basics for API calls	Intermediate
☐	Logging module — levels, handlers, formatters	Intermediate
☐	Unit testing — unittest, pytest basics	Intermediate
☐	Profiling & memory optimization — cProfile, memory_profiler	Advanced
☐	Multiprocessing vs multithreading — GIL awareness	Advanced
☐	Python packaging — setup.py, pyproject.toml	Advanced

■ 2. DATA MANIPULATION & MATH

NumPy

☐	Array creation — np.array, zeros, ones, arange, linspace	Beginner
☐	Array shapes — reshape, flatten, squeeze, expand_dims	Beginner
☐	Indexing & slicing — basic, fancy, boolean masking	Beginner
☐	Broadcasting rules	Intermediate
☐	Universal functions (ufuncs) — np.sum, mean, std, max, min	Beginner
☐	Matrix operations — dot product, matmul, transpose	Intermediate
☐	Linear algebra — np.linalg (eig, svd, inv, norm)	Intermediate
☐	Random module — np.random.seed, randn, randint, choice	Beginner
☐	Stacking — vstack, hstack, concatenate	Intermediate
☐	Performance: vectorization vs Python loops	Intermediate

Pandas

☐	Series & DataFrame creation from dict, list, CSV, JSON	Beginner
☐	Selecting data — loc, iloc, boolean filtering	Beginner
☐	Handling missing values — isnull, fillna, dropna	Beginner
☐	Data types — dtypes, astype, pd.to_datetime	Beginner
☐	GroupBy — groupby, agg, transform, apply	Intermediate
☐	Merging / joining — merge, join, concat	Intermediate
☐	Pivot tables & crosstabs	Intermediate
☐	String operations — str accessor methods	Intermediate
☐	Datetime operations — dt accessor, resample, rolling	Intermediate
☐	Apply & map — row/column-wise transformations	Intermediate
☐	MultiIndex DataFrames	Intermediate
☐	Memory optimization — categoricals, chunked reading	Advanced
☐	Method chaining & pipe for clean pipelines	Advanced

Math Foundations for ML		
☐	Linear Algebra — vectors, matrices, dot products, norms	Beginner
☐	Statistics — mean, median, mode, variance, std dev	Beginner
☐	Probability basics — conditional prob, Bayes theorem	Intermediate
☐	Distributions — Normal, Bernoulli, Binomial, Poisson	Intermediate
☐	Calculus intuition — gradients, chain rule (for backprop)	Intermediate
☐	Eigenvalues & eigenvectors (for PCA)	Intermediate
☐	Information theory — entropy, cross-entropy, KL divergence	Advanced

■ 3. MACHINE LEARNING (Classical)

Core Concepts & Workflow		
☐	Supervised vs Unsupervised vs Reinforcement Learning	Beginner
☐	Train / Validation / Test split strategy	Beginner
☐	Overfitting vs Underfitting — bias-variance tradeoff	Beginner
☐	Cross-validation — k-fold, stratified k-fold	Intermediate
☐	Feature engineering — encoding, scaling, binning	Intermediate
☐	Feature selection — correlation, mutual information, RFE	Intermediate
☐	Handling imbalanced datasets — SMOTE, class weights, undersampling	Intermediate
☐	Data leakage — what it is and how to avoid it	Intermediate
☐	Scikit-learn pipeline (Pipeline, ColumnTransformer)	Intermediate
☐	Experiment tracking — MLflow basics	Advanced

Algorithms		
☐	Linear Regression — OLS, assumptions, R2, RMSE	Beginner
☐	Logistic Regression — log-odds, sigmoid, decision boundary	Beginner
☐	Decision Trees — CART, Gini, information gain, pruning	Beginner
☐	Random Forest — bagging, feature importance	Intermediate
☐	Gradient Boosting — XGBoost, LightGBM (how they differ)	Intermediate
☐	Support Vector Machines — kernel trick, C, gamma	Intermediate
☐	Naive Bayes — text classification use case	Beginner
☐	K-Nearest Neighbors — distance metrics, curse of dimensionality	Beginner
☐	K-Means Clustering — elbow method, silhouette score	Intermediate
☐	PCA — dimensionality reduction, explained variance ratio	Intermediate
☐	DBSCAN & hierarchical clustering	Advanced
☐	Anomaly detection — IsolationForest, Z-score methods	Intermediate

Evaluation Metrics

☐	Classification — accuracy, precision, recall, F1, ROC-AUC	Beginner
☐	Confusion matrix interpretation	Beginner
☐	Regression — MAE, MSE, RMSE, R2, MAPE	Beginner
☐	Precision-Recall curve vs ROC curve — when to use which	Intermediate
☐	Log-loss / cross-entropy loss	Intermediate
☐	Multi-class metrics — macro, micro, weighted averages	Intermediate
☐	Calibration of probability estimates	Advanced

■ 4. DEEP LEARNING & NLP

Neural Networks Fundamentals

☐	Perceptron → MLP — layers, neurons, activation functions	Beginner
☐	Forward pass & backpropagation intuition	Beginner
☐	Activation functions — ReLU, Sigmoid, Tanh, Softmax, GELU	Beginner
☐	Loss functions — MSE, cross-entropy, binary cross-entropy	Beginner
☐	Optimizers — SGD, Adam, AdaGrad, RMSProp	Intermediate
☐	Batch size, learning rate, epochs — practical tuning	Intermediate
☐	Regularization — Dropout, L1/L2, Batch Normalization	Intermediate
☐	Weight initialization — Xavier, He	Intermediate
☐	PyTorch basics — tensors, autograd, nn.Module, DataLoader	Intermediate
☐	Early stopping & learning rate scheduling	Intermediate

NLP Foundations

☐	Text preprocessing — tokenization, lowercasing, stopwords, stemming	Beginner
☐	Bag of Words & TF-IDF representations	Beginner
☐	Word embeddings — Word2Vec, GloVe intuition	Intermediate
☐	Recurrent networks — RNN, LSTM, GRU (concept level)	Intermediate
☐	Attention mechanism — query, key, value intuition	Intermediate
☐	Transformer architecture — encoder, decoder, self-attention	Intermediate
☐	Tokenizers — BPE, WordPiece (used in BERT/GPT)	Intermediate
☐	Positional encoding	Intermediate
☐	Named Entity Recognition (NER), POS tagging basics	Beginner
☐	Text classification pipeline end-to-end	Beginner

■ 5. GENERATIVE AI, LLMs & RAG

LLM Fundamentals

☐	GPT architecture — decoder-only transformer, causal attention	Beginner
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☐	BERT architecture — encoder-only, bidirectional attention	Beginner
☐	Pre-training vs fine-tuning vs in-context learning	Beginner
☐	Prompt engineering — zero-shot, few-shot, chain-of-thought	Beginner
☐	System prompt, user prompt, assistant turn structure	Beginner
☐	Temperature, top-p, top-k, max_tokens parameters	Beginner
☐	Hallucination — why it happens, mitigation strategies	Intermediate
☐	Instruction tuning & RLHF overview	Intermediate
☐	PEFT methods — LoRA, QLoRA (concept + when to use)	Intermediate
☐	Context window limitations & chunking strategies	Intermediate
☐	Quantization — INT8, INT4, GGUF formats	Intermediate
☐	Model evaluation — BLEU, ROUGE, BERTScore, LLM-as-judge	Intermediate
☐	Structured outputs / JSON mode from LLMs	Intermediate
☐	Function calling / tool use in LLMs	Intermediate
☐	Multi-modal LLMs — vision + text (GPT-4V, Gemini Vision)	Advanced

RAG (Retrieval Augmented Generation) — YOUR CORE SKILL

■ You already built RAG at TCSI! Focus on depth: chunking strategies, re-ranking, evaluation.

☐	RAG pipeline architecture — ingest → embed → store → retrieve → generate	Beginner
☐	Document loaders — PDF, HTML, DOCX, web scraping	Beginner
☐	Text chunking strategies — fixed, recursive, semantic, sentence	Intermediate
☐	Chunk overlap — why it matters, optimal settings	Intermediate
☐	Embedding models — OpenAI, Vertex AI, sentence-transformers	Beginner
☐	Vector databases — FAISS (in-memory), ChromaDB, Pinecone, Weaviate	Intermediate
☐	Similarity search — cosine similarity, dot product, Euclidean	Beginner
☐	ANN indexes — HNSW, IVF (how FAISS scales)	Intermediate
☐	Hybrid search — combining dense + sparse (BM25) retrieval	Intermediate
☐	Re-ranking — cross-encoder re-rankers (Cohere, sentence-transformers)	Intermediate
☐	Metadata filtering in vector DBs	Intermediate
☐	Parent-document retrieval & small-to-big chunking	Intermediate
☐	HyDE (Hypothetical Document Embeddings)	Advanced
☐	RAG evaluation — faithfulness, answer relevancy, context precision (RAGAS)	Intermediate
☐	GraphRAG — entity-based retrieval using knowledge graphs	Advanced
☐	Multi-hop RAG for complex question answering	Advanced
☐	RAG vs fine-tuning — when to use which	Intermediate

LLM Frameworks

☐	LangChain — chains, LCEL syntax, runnable interface	Intermediate
☐	LangChain — retrievers, memory, callbacks	Intermediate

☐	LangChain — document loaders & text splitters	Intermediate
☐	LangSmith — tracing, evaluation, prompt management	Intermediate
☐	LlamaIndex — index types, query engine, response synthesizer	Intermediate
☐	Haystack — pipeline concept, nodes, evaluation	Intermediate
☐	Google Vertex AI SDK — generate_content, embeddings API	Beginner
☐	OpenAI SDK — chat completions, embeddings, function calling	Beginner
☐	Anthropic Claude API — messages format, streaming	Intermediate
☐	Semantic Kernel (Microsoft) — basics	Advanced

■ 6. AGENTIC AI & MULTI-AGENT SYSTEMS

Agent Fundamentals

☐	Agent loop — observe → think → act → observe (ReAct pattern)	Beginner
☐	Tools / Function calling — registering and handling tool results	Beginner
☐	Planning strategies — Chain-of-Thought, Tree-of-Thoughts, ReAct	Intermediate
☐	Memory types — in-context, external (vector DB), episodic, semantic	Intermediate
☐	Agent state management across turns	Intermediate
☐	Guardrails & safety in agents — output validation, human-in-the-loop	Intermediate
☐	Structured output parsing — Pydantic models for tool inputs/outputs	Intermediate

Google ADK — YOUR CURRENT STACK

■ Google ADK is your differentiator. Go deep here — most candidates don't have this.

☐	ADK agent definition — BaseAgent, LlmAgent structure	Beginner
☐	Tools in ADK — FunctionTool, built-in tools	Beginner
☐	Session management & state persistence in ADK	Intermediate
☐	Multi-agent patterns in ADK — coordinator + sub-agents	Intermediate
☐	Streaming responses in ADK agents	Intermediate
☐	Deploying ADK agents to Vertex AI Agent Engine	Intermediate
☐	Evaluation framework in ADK	Intermediate
☐	ADK + Gemini models — model selection, safety settings	Beginner
☐	ADK callbacks — before/after model, before/after tool	Advanced
☐	ADK custom authentication for tools	Advanced

Model Context Protocol (MCP)

☐	MCP architecture — host, client, server roles	Beginner
☐	MCP primitives — tools, resources, prompts	Beginner
☐	Building an MCP server in Python (FastMCP / official SDK)	Intermediate
☐	MCP tool schema — JSON schema, annotations	Intermediate

☐	Connecting MCP servers to Claude Desktop / ADK	Intermediate
☐	MCP resources — URI templates, list/read resource handlers	Intermediate
☐	MCP authentication & transport — SSE vs stdio	Intermediate
☐	MCP sampling — allowing server to request LLM completions	Advanced
☐	Production MCP server — error handling, logging, versioning	Advanced

Multi-Agent Frameworks

☐	LangGraph — nodes, edges, state graph, conditional routing	Intermediate
☐	LangGraph — human-in-the-loop, persistence, streaming	Intermediate
☐	CrewAI — agents, tasks, crew, process (sequential/hierarchical)	Intermediate
☐	AutoGen — conversable agents, group chat manager	Intermediate
☐	Agent communication patterns — supervisor, peer-to-peer, blackboard	Intermediate
☐	Error recovery in multi-agent systems	Advanced
☐	Agent observability — LangSmith, Langfuse tracing	Intermediate

■ 7. FASTAPI & BACKEND ENGINEERING

FastAPI Core — YOUR CURRENT STACK

☐	Route decorators — GET, POST, PUT, DELETE	Beginner
☐	Path parameters, query parameters, request body	Beginner
☐	Pydantic models for request/response validation	Beginner
☐	Dependency injection — Depends()	Intermediate
☐	Background tasks — BackgroundTasks	Intermediate
☐	Async endpoints — async def, await	Intermediate
☐	Middleware — CORS, logging, rate limiting	Intermediate
☐	Error handling — HTTPException, custom exception handlers	Intermediate
☐	File uploads — UploadFile, File	Intermediate
☐	Streaming responses — StreamingResponse (for LLM token streaming)	Intermediate
☐	WebSockets in FastAPI	Intermediate
☐	Authentication — JWT tokens, OAuth2 with FastAPI	Intermediate
☐	OpenAPI docs — tags, descriptions, response models	Beginner
☐	Testing — TestClient, pytest fixtures for FastAPI	Intermediate
☐	Lifespan events — startup/shutdown (DB connections, model loading)	Intermediate

API Design & System Design Basics

☐	REST principles — statelessness, resource naming conventions	Beginner
☐	HTTP status codes — 200, 201, 400, 401, 403, 404, 422, 500	Beginner
☐	Pagination — offset-based, cursor-based	Intermediate

☐	Rate limiting concepts	Intermediate
☐	Caching — Redis basics for API response caching	Intermediate
☐	Message queues — Pub/Sub, Cloud Tasks basics	Intermediate
☐	Microservices vs monolith tradeoffs	Intermediate
☐	API versioning strategies	Intermediate
☐	System design components — load balancer, DB, cache, queue	Advanced

■ ■ 8. GOOGLE CLOUD PLATFORM & MLOps

GCP Core Services — YOUR CURRENT STACK

☐	IAM — roles, service accounts, principle of least privilege	Beginner
☐	Cloud Storage — buckets, object storage, signed URLs, lifecycle	Beginner
☐	BigQuery — dataset/table structure, basic SQL queries	Beginner
☐	BigQuery — partitioning, clustering for performance	Intermediate
☐	BigQuery ML (BQML) — training models in SQL	Intermediate
☐	Cloud Run — deploying containers, environment variables, scaling	Beginner
☐	Cloud Run — concurrency, min instances, CPU allocation	Intermediate
☐	Artifact Registry — storing Docker images	Intermediate
☐	Cloud Build — CI/CD pipelines for deployment	Intermediate
☐	Secret Manager — storing API keys securely	Intermediate
☐	Pub/Sub — topics, subscriptions, push vs pull	Intermediate
☐	Cloud Logging & Cloud Monitoring — alerts, dashboards	Intermediate

Vertex AI — YOUR CURRENT STACK

☐	Vertex AI Workbench — managed Jupyter notebooks	Beginner
☐	Vertex AI Datasets — creating & managing datasets	Beginner
☐	Vertex AI Training — custom training jobs, pre-built containers	Intermediate
☐	Vertex AI Model Registry — versioning, aliases	Intermediate
☐	Vertex AI Endpoints — deploying models, traffic splitting	Intermediate
☐	Vertex AI Pipelines — Kubeflow Pipelines on Vertex	Intermediate
☐	Vertex AI Feature Store	Intermediate
☐	Vertex AI Agent Builder — search & conversation apps	Intermediate
☐	Vertex AI Agent Engine — deploying ADK agents	Intermediate
☐	Gemini API on Vertex — models, safety settings, grounding	Beginner
☐	Embeddings API on Vertex — text-embedding-004 & multimodal	Beginner
☐	Vertex AI Vector Search (Matching Engine)	Intermediate

Containerization & DevOps

☐	Docker basics — Dockerfile, build, run, tag, push	Beginner
☐	Docker Compose — multi-service local development	Beginner
☐	Docker best practices — multi-stage builds, .dockerignore	Intermediate
☐	Container registries — Artifact Registry, Docker Hub	Beginner
☐	Kubernetes concepts — pods, deployments, services, ingress	Intermediate
☐	GKE (Google Kubernetes Engine) basics	Intermediate
☐	CI/CD concepts — Git workflow, automated testing, deployment	Intermediate
☐	Infrastructure as Code — Terraform basics	Advanced

MLOps Practices

☐	ML pipeline stages — data validation, training, evaluation, serving	Intermediate
☐	Model versioning & lineage tracking	Intermediate
☐	Model monitoring — data drift, concept drift detection	Intermediate
☐	A/B testing for model evaluation in production	Intermediate
☐	MLflow — experiment tracking, model registry	Intermediate
☐	Feature stores — purpose, offline vs online serving	Intermediate
☐	Continuous training triggers	Advanced
☐	Model explainability — SHAP values, Vertex Explainable AI	Advanced

9. SQL & DATA ENGINEERING

SQL Fundamentals to Advanced

☐	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT	Beginner
☐	Aggregate functions — COUNT, SUM, AVG, MIN, MAX	Beginner
☐	GROUP BY, HAVING	Beginner
☐	JOINS — INNER, LEFT, RIGHT, FULL OUTER, CROSS	Beginner
☐	Subqueries & CTEs (WITH clause)	Intermediate
☐	Window functions — ROW_NUMBER, RANK, LAG, LEAD, PARTITION BY	Intermediate
☐	String functions — CONCAT, SUBSTR, REPLACE, REGEXP	Intermediate
☐	Date/time functions — DATE_DIFF, DATE_TRUNC, EXTRACT	Intermediate
☐	CASE WHEN expressions	Intermediate
☐	BigQuery-specific — UNNEST arrays, STRUCT, JSON functions	Intermediate
☐	Query optimization — EXPLAIN, index usage, partitioning	Advanced
☐	Data modeling — star schema, fact vs dimension tables	Advanced

10. DSA & CODING INTERVIEWS

■ For AI/ML roles at product companies (Google, Microsoft), DSA is tested. For AI-focused startups & TCS-level companies, practical coding matters more.

Data Structures

☐	Arrays & strings — two pointers, sliding window patterns	Beginner
☐	Hash maps & hash sets — frequency counting, lookups	Beginner
☐	Linked lists — traversal, reversal, fast/slow pointers	Beginner
☐	Stacks & queues — implementation, use cases	Beginner
☐	Binary search — on sorted array and answer space	Intermediate
☐	Trees — traversal (BFS/DFS), height, lowest common ancestor	Intermediate
☐	Heaps/Priority queues — top-K problems	Intermediate
☐	Graphs — BFS, DFS, topological sort, Union-Find	Intermediate
☐	Dynamic programming — memoization, tabulation, common patterns	Advanced

Complexity & Problem Solving

☐	Big-O time & space complexity analysis	Beginner
☐	Recognizing patterns — two pointers, sliding window, fast/slow ptr	Intermediate
☐	Greedy algorithms	Intermediate
☐	Backtracking — permutations, combinations, N-queens	Advanced
☐	Bit manipulation basics	Intermediate
☐	LeetCode Easy — solve 30+ problems across key topics	Beginner
☐	LeetCode Medium — solve 20+ problems, especially arrays/trees/graphs	Intermediate

11. PROJECTS, PORTFOLIO & COMMUNICATION

Portfolio Projects (Build These)

☐	Production-grade RAG app with evaluation (RAGAS metrics) ■ Use LangChain/LlamaIndex + FAISS/ChromaDB + FastAPI	Intermediate
☐	LangGraph or CrewAI multi-agent system with a real use case ■ E.g. research agent, code review agent	Intermediate
☐	MCP server for a useful tool (calendar, DB, web search) ■ Show ADK + MCP integration	Intermediate
☐	ML classification project with full pipeline on Vertex AI ■ End-to-end: EDA → training → deployment → monitoring	Intermediate
☐	Open-source contribution (even docs/examples) to ADK or LangChain	Advanced
☐	Technical blog post or LinkedIn article on a project you built	Beginner
☐	GitHub profile — pinned repos, README, clean commits	Beginner

Communication & Interview Skills

☐	STAR method for behavioral questions (Situation, Task, Action, Result)	Beginner
☐	Ability to explain ML/AI concepts to non-technical stakeholders	Beginner

☐	System design of AI systems — articulate RAG/agent architecture clearly	Intermediate
☐	Writing clear technical documentation & ADRs	Intermediate
☐	Presenting ML results — metrics, tradeoffs, business impact	Intermediate

■ 2-WEEK SKILL AUDIT & LEARNING PLAN

■ Goal of Week 1: Audit everything you know. Fill checkboxes honestly. Identify real gaps. Goal of Week 2: Fill the most critical gaps with focused 2-3 hour daily sessions.

Day 1	Python Audit	Work through Python sections 1.1–1.3. Write small scripts to test each concept you're unsure about. No tutorials — just code.
Day 2	NumPy & Pandas Audit	Load a real CSV (your ticket data if possible). Perform groupby, merge, pivot, apply operations. Check boxes as you do each.
Day 3	ML Concepts Audit	Pick a classification dataset (UCI/Kaggle). Build: preprocessing pipeline → model → evaluation. Document gaps.
Day 4	RAG Deep Dive	Re-build your TCS RAG pipeline from scratch locally. Add chunking strategy comparison. Add RAGAS evaluation. Time: 3 hrs.
Day 5	ADK + MCP Deep Dive	Build a fresh ADK agent with a custom MCP server. Document what you understand vs what you had to look up.
Day 6	FastAPI + GCP	Deploy a FastAPI endpoint to Cloud Run from scratch. Add auth, logging, error handling. Time yourself.
Day 7	REVIEW & GAP LIST	Look at all unchecked boxes. Write a prioritized gap list: Top 10 things to learn in Week 2.
Day 8	Fill Gap #1-2	Focus on your 2 highest-priority gaps. Deep study: docs + coding exercise per topic. No videos — hands-on only.
Day 9	Fill Gap #3-4	Same approach. If DSA is a gap: solve 5 LeetCode Easy + 3 Medium problems.
Day 10	LLM Internals	Study transformer architecture (Illustrated Transformer blog). Implement attention from scratch (small version). Solidify.
Day 11	MLOps & Vertex AI	Walk through a Vertex AI Pipeline end-to-end. Add model monitoring. Check those boxes.
Day 12	System Design Practice	Design a RAG system for 1M documents on paper. Then design a multi-agent ticket classifier. Practice explaining aloud.
Day 13	Mock Interview Day	Do 2 mock technical interviews (use Pramp or a friend). 1 coding round + 1 ML concept round.
Day 14	Resume & Portfolio Update	Update resume with confidence. Push 2 clean projects to GitHub. Write 1 LinkedIn post about a project. Apply to 3 target companies.

■ TARGET ROLES FOR COMPANY SWITCH (2025-2026 Market)

Role	Fit Score	Why This Fits You	Salary Range (India)	What to Build/Learn
AI/ML Engineer (GenAI Focused)	★★★★ ★ Best Fit	Your #1 target. 2 years TCS GenAI + RAG + ADK + MCP + Vertex AI is exactly what companies want.	■18–32 LPA (India) \$130–180K (US/remote)	Deepen LLM internals, RAG evaluation, LangGraph. Build 2 strong portfolio projects.

MLOps / AI Platform Engineer	★★★★ ■ <i>Strong Fit</i>	Your deployment skills (Cloud Run, Vertex AI, FastAPI) are relevant. Gap: CI/CD pipelines, Kubeflow.	■18–28 LPA	Learn Vertex AI Pipelines, Kubeflow, model monitoring deeply.
Data Scientist	★★★ ■ <i>Moderate Fit</i>	You'll need stronger stats, EDA, and classical ML depth. Your AI engineering skills are overkill for pure DS.	■12–22 LPA	Fill stats gaps, practice EDA storytelling, learn visualization (matplotlib, seaborn, plotly).
AI Research Engineer	★★ ■ <i>Future Target</i>	Requires deep ML theory, paper implementation skills. Target after 1-2 more years of depth.	■25–45 LPA	Read ML papers, implement models from scratch, contribute to open source ML.
Backend Engineer (AI Systems)	★★★★ ■ <i>Solid Option</i>	Your FastAPI + GCP + Python skills qualify. Position yourself as 'AI-aware backend engineer'.	■15–25 LPA	Strengthen system design, distributed systems basics, database design.

■ A NOTE ON CONFIDENCE

Dinesh — confidence doesn't come from knowing everything. It comes from knowing *where you stand*. The impostor syndrome you feel is extremely common among engineers who build real things (vs those who only watch tutorials). You have deployed RAG pipelines and ADK agents in production at a major company. That is real. Use this checklist not to feel bad about gaps, but to convert unknown unknowns into known unknowns — that alone will double your confidence. Check each box only when you can **explain it to someone else**, not just when you've read about it. That standard will make you genuinely strong.